$$
\begin{aligned}
l_t^e &= g_e(\widetilde{o}_t^e) \in \mathbb{R}^{96} \\
(b'_t, h_{t+1}) &= \mathrm{GRU}(o_t^p, l_t^e, h_t) \\
\alpha_t &= \sigma(g_a(b'_t)) \in (0,1)^{96} \\
b_t &= g_b(b'_t) + \mathrm{pad}_{120}(l_t^e \odot \alpha_t) \\
a_t^S &= \pi_S([o_t^p, b_t]) \in \mathbb{R}^{16}
\end{aligned}
$$